THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 7: **Coordinate Geometry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The distance between $(2, 3)$ and $(5, 7)$ is: 1

(a) 7 **(b)** 4 **(c)** 6 **(d)** 5

2. The point which divides the join of $(0, 0)$ and $(6, 9)$ in the ratio 1:2 internally is: 1

(a) $(6,\ 9)$ **(b)** $(4,\ 6)$ **(c)** $(2,\ 3)$ **(d)** $(3,\ \frac{9}{2})$

3. The area of the triangle with vertices $(0, 0)$, $(4, 0)$ and $(0, 6)$ is: 1

(a) 13 **(b)** 14 **(c)** 12 **(d)** 11

4. If the midpoint of $(2, 3)$ and (x, y) is $(5, 6)$, then $x =$ 1

(a) 7 **(b)** 9 **(c)** 10 **(d)** 8

5. The distance of $(5, 3)$ from the y-axis is: 1

(a) 5 **(b)** 7 **(c)** 4 **(d)** 6

6. The point $(0, 0)$ is equidistant from $(2, 3)$ and $(6, 1)$ for $x =$ 1

(a) 2 **(b)** 3 **(c)** 5 **(d)** 4

7. The distance between $(-1, 2)$ and $(4, 14)$ is: 1

(a) 14 **(b)** 13 **(c)** 12 **(d)** 15

8. The point which divides the join of $(3, 3)$ and $(9, 12)$ in the ratio 1:2 internally is: 1

(a) $(7,\ 9)$ **(b)** $(5,\ 6)$ **(c)** $(9,\ 12)$ **(d)** $(6,\ \frac{15}{2})$

9. The area of the triangle with vertices $(0, 0)$, $(8, 0)$ and $(0, 5)$ is: 1

(a) 22 **(b)** 20 **(c)** 19 **(d)** 21

10. If the midpoint of $(4, 1)$ and (x, y) is $(7, 2)$, then $x =$ 1

(a) 9 **(b)** 10 **(c)** 11 **(d)** 12

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the distance between the points $(2,\ 3)$ and $(4,\ 1)$. 2

12. Find the coordinates of the point which divides the join of $(4,\ -3)$ and $(8,\ 5)$ internally in the ratio $3 : 1$. 2

13. Find the values of y for which the distance between $P(10,\ y)$ and $Q(2,\ -3)$ is 10 units. 2

14. Find the distance between the points $(-5, 7)$ and $(-1, 3)$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the value of k for which the points $(1, 2)$, $(3, 6)$ and $(5, k)$ are collinear. 3

16. Find the area of the triangle with vertices $(1, 1)$, $(4, 5)$ and $(7, 2)$. 3

17. In what ratio does the x-axis divide the segment joining $(-4, -6)$ and $(3, 8)$? Find the point of division. 3

OR

Find the value of k for which the points $(2, 1)$, $(4, 5)$ and $(6, k)$ are collinear. 3

18. Find the area of the triangle with vertices $(2, 3)$, $(-1, 0)$ and $(2, -4)$. 3

OR

In what ratio does the x-axis divide the segment joining $(2, -3)$ and $(6, 9)$? Find the point of division. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Find the point on the x-axis which is equidistant from $(7, 6)$ and $(-3, 4)$. 5

OR

Three vertices of a parallelogram $ABCD$ taken in order are $A(1, 2)$, $B(4, 3)$ and $C(5, 7)$. Find the fourth vertex D. 5

20. Find the point on the x-axis which is equidistant from $(2, 3)$ and $(8, 1)$. 5

OR

Three vertices of a parallelogram $ABCD$ taken in order are $A(2, 1)$, $B(6, 2)$ and $C(7, 6)$. Find the fourth vertex D. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND